LYLE OKOTH

DATA ENGINEER | AUDIO & ML DATA PIPELINES

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PROFESSIONAL SUMMARY

Data Engineer with 6+ years of experience building batch and streaming pipelines that turn raw, noisy audio into training-ready datasets at scale — 6 TB / 38,000 hours per year — for machine learning teams. Owns ingestion, alignment, labelling, versioning and data quality end to end, with hands-on software engineering (APIs at 1.2M requests/day) and ML experience that keeps pipelines tuned for model training, not just analytics. Cut human data-labelling time 68% through automation and reduced pipeline failures 41% by consolidating 14 scattered schemas into a dbt-modelled warehouse.

CORE COMPETENCIES

Pipeline Engineering: Batch ETL / ELT, streaming & event-driven jobs, Spark tuning for audio shards, Parquet/Arrow data lakes, feature stores, orchestration with Airflow and Dagster-style dependency graphs

Data Quality & Governance: Great Expectations checks, dataset versioning (DVC) + content hashing, schema consolidation with dbt, speaker consent, retention and PII-redaction flows for 40,000+ hours of stored audio

ML Data Support: Transcription alignment, diarisation pre-labelling, human + AI collaborative annotation platforms, synthetic data generation, data pipelines tuned for ASR/TTS/translation training

Platform & Reliability: AWS S3, GCS, PostgreSQL, Elasticsearch, Redis, Kafka, Terraform, Docker, on-call and observability: moved from monthly restores to daily automated checks

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, C++

Processing: Apache Spark, Kafka, Airflow, dbt, DuckDB, Flink-style streaming, Celery

Storage & Warehousing: PostgreSQL, AWS S3, GCS, Parquet / Arrow, Elasticsearch, Redis

Quality & Versioning: Great Expectations, DVC, content hashing, data contracts

Infrastructure: Docker, Kubernetes, Terraform, GitHub Actions CI/CD, AWS, GCP

PROFESSIONAL EXPERIENCE

Data Engineer — SautiFlow Labs, Nairobi 2022 – 2024

- Designed batch and streaming pipelines with Apache Spark, Airflow and Kafka processing 6 TB / 38,000 hours of raw audio per year into training-ready datasets for ASR, translation and TTS models.
- Automated transcription alignment and diarisation pre-labelling, cutting human labelling time from 11 to 3.5 minutes per hour of audio — a 68% saving adopted as the platform's default workflow.
- Introduced dataset versioning with DVC and content hashing so every model release traces to an exact data snapshot, eliminating reproducibility disputes across the ML team.
- Consolidated 14 scattered schemas into a single dbt-modelled warehouse, reducing pipeline failures by 41% and monthly compute spend by 23%.
- Built data-quality runbooks (Great Expectations) and onboarded 3 team members; owned speaker consent, retention and PII-redaction flows for 40,000+ hours of stored audio.
- Engineered feature stores and Parquet data lakes tuned for audio shards rather than tabular rows; moved pipeline reliability from monthly restores to daily automated checks.

Machine Learning Engineer — SautiFlow Labs, Nairobi 2024 – Present

- Own the speech model lifecycle end to end; the data foundation built in this role feeds all model training — 1,240 hours of curated audio drove a Whisper ASR fine-tune that cut WER from 23.4% to 11.8%.
- Built the audio data collection, editing and annotation platform leveraging fine-tuned models for AI-assisted labelling; published 3 open datasets to Hugging Face (18,000+ downloads).

Software Engineer — SautiFlow Labs, Nairobi 2020 – 2022

- Built and maintained REST and WebSocket APIs in Django and Node.js serving 1.2M requests/day at p95 latency under 140 ms, with PostgreSQL, Redis and Twilio voice/SMS integrations.
- Introduced Docker deployments and a GitHub Actions CI/CD pipeline: release cadence moved from monthly to weekly and lead time dropped from 3 weeks to 2 days.

SELECTED PROJECTS

LabelSet — Speech Data Pipeline — Sources recordings, gets them labelled by real speakers, and verifies label quality before training — the data backbone behind every model in the portfolio.

EchoID — Audio Archive Indexing — Batch pipeline applying diarisation, voice embeddings and ASR to turn raw archives into speaker-labelled, searchable transcripts.

SautiFlow — Script-to-Skit Product — Turns a written script into a finished skit — casting, voicing, timing and mix — giving creators ten renderable variations per joke; served end to end by the audio ingestion, alignment and labelling pipelines described above.

EDUCATION

BSc, Computer Security and Forensics — Jaramogi Oginga Odinga University of Science and Technology (JOOUST) 2016 – 2020

Diploma, Mechatronics Engineering — JKUAT, Juja, Kenya 2013 – 2016

CERTIFICATIONS

AWS, Google Cloud, DeepLearning.AI and NVIDIA DLI certifications (2021 – 2025).

REFERENCES

Available on request.